The examination is being carried out on the following application documents:

Description, Pages	1-21	filed with entry into the regional phase before the EPO
Claims, Numbers	1-12	filed with entry into the regional phase before the EPO
Drawings, Sheets	1/3-3/3	filed with entry into the regional phase before the EPO

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure:

D1	CN 216 348 930 U (WEIFANG ENERGY GATHERING BATTERY LTD COMPANY) 19 April 2022
D2	CN 108 956 370 A (GUIZHOU ACAD OF TESTING AND ANALYSIS) 7 December 2018
D3	CN 203 490 042 U (XIAMEN CHL NEW ENERGY CO LTD) 19 March 2014

Notwithstanding the clarity objections raised in paragraph 3 of this European Search Opinion, the present application does not meet the requirements of the Article 52(1) EPC because the subject-matter of claims 1, 3 and 5 is not new (Article 54 (1) and (2) EPC).

The subject-matter of claims 2, 4, 6-9 and 11-12 does not meet the requirements of the EPC in respect of the inventive step (Article 56 EPC).

The subject-matter of claim 10 seems to meet the requirements of the EPC in respect of both novelty (Article 54 (1) and (2) EPC) and of the inventive step (Article 56 EPC).

1 Independent claim 1

1.1 Article 54 (1) and (2) EPC

D1 discloses all the features of independent claim 1 (references in parentheses applying to this document):

Claim 1: A gas volume measurement apparatus (**fig 1**) for monitoring volume of gas produced during formation of a battery (**"A lithium-ion battery formation gas production detection device" in claim 1**), comprising:
a delivery device (**1-6 in fig 1**) and a measurement device (**U tube 7 in fig 1**); wherein
the delivery device connects the battery (**claim 1**) and the measurement device (**7 in fig 1**) and is configured to remove hazardous substances from gas produced by the battery (**as tube 3 dives below the surface of water in bottle 4 it is considered that the gas is purified in the process, this feature is the same as in the claimed apparatus**) and deliver the gas to the measurement device (**6 in fig 1**); and
the measurement device (**7 in fig 1**) is configured to measure volume of the gas produced by the battery (**claim 1**).

Hence the subject-matter of **independent claim 1** is not new (Article 54(1) and (2) EPC)**.**

1.2 **Article 56 EPC**

Document D2, regarded as being the prior art closest to the subject-matter of claim 1, discloses (references in parentheses applying to this document):

Claim 1: A gas volume measurement apparatus suitable for monitoring the volume of gas produced during formation of a battery (**see in fig 1**), comprising:
a delivery device (**see 2, 4 and 5 in fig 1**) and a measurement device (**U tube 3 in fig 1**); wherein
the delivery device connects the ~~battery~~ hydrolysis reaction generator (**12 in fig 1**) and the measurement device (**3 in fig 1**) and is configured to remove hazardous substances from gas produced by the ~~battery~~ hydrolysis reaction generator (**"The gas purifying device 2 includes a gas scrubbing bottle 21" in § [0031]**) and deliver the gas to the measurement device (**see 5 in fig 1**); and
the measurement device is configured to measure volume of the gas produced by the ~~battery~~ hydrolysis reaction generator (**"The gas pressure detecting device 3 includes a U-shaped pressure gauge 31" in § [0032]**).

The subject-matter of claim 1 differs from the apparatus of document D2 in that the gas volume measurement device is to be connected to a battery instead of a hydrolysis reaction generator.

The technical effect of this difference is that the device to be monitored is a battery.

The objective technical problem to be solved may be regarded as how to use the same volume monitoring device on a battery.

When confronted with the above-mentioned problem, it would be obvious for the person skilled in the art, who is skilled in the analysis of materials by measuring the volume of vapours or gas emitted by a device, to apply the teaching of D2 to a battery and obtaining the subject-matter of claim 1.

The subject-matter of claim 1 does therefore not involve an inventive step (Article 56 EPC).

2 Dependent claims 2-12

The subject-matter of the following claims is considered as not inventive over the cited prior art.

2.1 Article 54 (1) and (2) EPC

D1 discloses all the features of the following dependent claims (references in parentheses applying to this document):

claims 3 and 5: see bottle 4 filled with water (last paragraph on page 6).

2.2 Article 56 EPC

D1 is considered to be the prior art closest to the subject-matter of the following claims:

Claim 2: D1 is concerned with providing accurate results (see 3rd paragraph on page 6); both processing devices and calibration / compensation steps are well known in the art. For the person skilled in the art, it would be an obvious design option to provide a processing device and calibration step.

claim 4: D1 does not disclose any buffer bottle. Yet it is known that the set-up of D1 might lead to some fluid of bottle 4 to be sucked back towards tube inlet 1, which can be avoided by the presence of a buffer bottle between the tube inlet and the water bottle. For example see in D3 (buffer bottle 2 in fig 1).

claim 6: see rubber stopper 5 in fig 1 in D1, as well as gel stopper 11 in fig 1 in D3.

claim 7: see tubes 3 and 6 in fig 1 in D1, as well tubes 3, 4 and 5 in fig 1 in D3.

claims 8 and 9: The feature of the capacities and height ranges is merely one of several straightforward possibilities which the skilled person would select, depending on the circumstances, without exercising inventive skill, in order to solve the problem of choosing proper dimensions for the bottles.

claims 11 and 12: following the reasoning for claim 2, for the person skilled in the art, it would be quite obvious to record the measured data and to display them in the form of an analytical curve to an operator, so that said operator might want to adjust some parameters of the battery formation in return.

2.3 The subject-matter of **claims 10** seems to meet the requirements of Article 56 EPC as it seems novel and inventive over the prior art documents for the following reasons :

claim 10: the equation for the pressure-temperature correction in claim 10 is not disclosed in neither of D1-D3, nor is it considered obvious.

3 Additional remarks

3.1 Article 84 EPC

3.1.1 There is insufficient antecedent basis for the term "buffer bottle" in the claim 6 as being dependent on claim 5. The buffer bottle is only introduced in claim 4, which claim 6 is not dependent on.

4 Final remarks

When filing amended claims, the applicant should at the same time:

- bring the independent claims in the two-part form in accordance with Rule 43(1) EPC, with those features known from the prior art being placed in the preamble (Rule 43(1)(a) EPC) and the remaining features being included in the characterizing part (Rule 43(1)(b) EPC),
- bring the description in conformity with the amended claims,
- identify document D1 in the description and its relevant contents, to meet the requirements of Rule 42(1)(b) EPC. The applicant should ensure that it is clear from the description which features of the subject-matter of independent claim(s) are known from D1.

The attention of the applicant is drawn to the fact that the application may not be amended in such a way that it contains subject-matter which extends beyond the content of the application as filed (Article 123(2) EPC).

In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments in addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based (see Guidelines H-III,2.2). If the applicant considers it appropriate, these indications could be submitted in handwritten form on a copy of the relevant parts of the application as filed or in documents with tracked changes.